Function Notation: When $f(x)$ Is Not Multiplication

Read the notation before you compute:

- $f(x)$ is the *output* of the function f at the input x. The parentheses hold an input slot, not a factor.
- $f(3)$ means "replace every x in the rule with 3." It does not mean f times 3.
- f on its own is not a number, so nothing can be distributed across it.
- $2f(3)$ doubles the *output*; $f(2 \cdot 3)$ doubles the *input*. These are different instructions.

For every find-and-fix item, quote the exact step where the notation was misread, then compute the correct value.

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1. Let $f(x) = x + 4$. Find $f(3)$.
Write the substitution step before the arithmetic.

Answer: ____________________

2. Let $g(x) = x^2 - 2$. Find $g(5)$.

Answer: ____________________

3. Let $f(x) = x^2 + 1$. Compute $2f(3)$.
Then state in one line how $2f(3)$ differs from $f(2 \cdot 3)$.

Answer: ____________________

4. Find and fix the mistake. Leo is given $f(x) = 4x - 1$ and asked for $f(3)$. He writes $f(3) = 3(4x - 1)$ and reports 33.
Name the misreading, then give the correct value of $f(3)$.

Answer: ________________

5. Find and fix the mistake. Sam is given $f(x) = x^2 + 3x$ and asked for $f(5)$. He writes $f(2 + 3) = f(2) + f(3)$ and reports 28.
Explain which property Sam borrowed that functions do not obey in general, then give the correct value of $f(5)$.

Answer: ________________

6. Let $f(x) = 5x - 12$. Find every input x for which $f(x) = 8$.
Say in one phrase why this is an equation to solve and not a product to divide.

Answer: ________________

7. Let $h(x) = x^2 - 4x$. Find every input x for which $h(x) = 0$.
A classmate reads $h(x) = 0$ as "h times x equals 0" and answers $x = 0$ only. Show what that reading misses.

Answer: ________________

8. **Explain.** Write a short paragraph a classmate could use to tell the two readings of $f(x)$ apart: name what the parentheses hold, say why f by itself cannot be multiplied by anything, and give one concrete test that exposes the multiplication misreading.